

HK IMO Training 2020-2021
Problem Set 14
For 6 February 2021



Problem 53. In triangle ABC , let I be the incentre. Denote the circumcircles of $\triangle ABC$ and $\triangle BIC$ by Ω and ω respectively. In Ω , let M be the midpoint of arc BC not containing A , and let N be the midpoint of arc BC containing A . Suppose NI meets ω the second time at P . Prove that the midpoint of IP lies on Ω .

Problem 54. Let I be the incenter of a triangle ABC , M be the midpoint of AC , and W be the midpoint of arc AB of the circumcircle not containing C . It is known that $\angle AIM = 90^\circ$. Find the ratio $CI : IW$.

Problem 55. In triangle ABC , let I and O be the incentre and circumcentre respectively. Suppose the incircle and A -excircle touch BC at D and E respectively. Let M be the second intersection of AI and the circumcircle of $\triangle ABC$, and let S be the intersection of DM and OI . Prove that $\angle BAS = \angle EAC$.

Problem 56. Let Ω be the circumcircle of the triangle ABC . Two circles ω_1, ω_2 are tangent to each of the circle Ω and the rays AB, AC , with ω_1 interior to Ω , ω_2 exterior to Ω . The common tangent of Ω, ω_1 and the common tangent of Ω, ω_2 intersect at the point X . Let M be the midpoint of the arc BC (not containing the point A) on the circle Ω , and the segment AA' be the diameter of Ω . Prove that X, M , and A' are collinear.